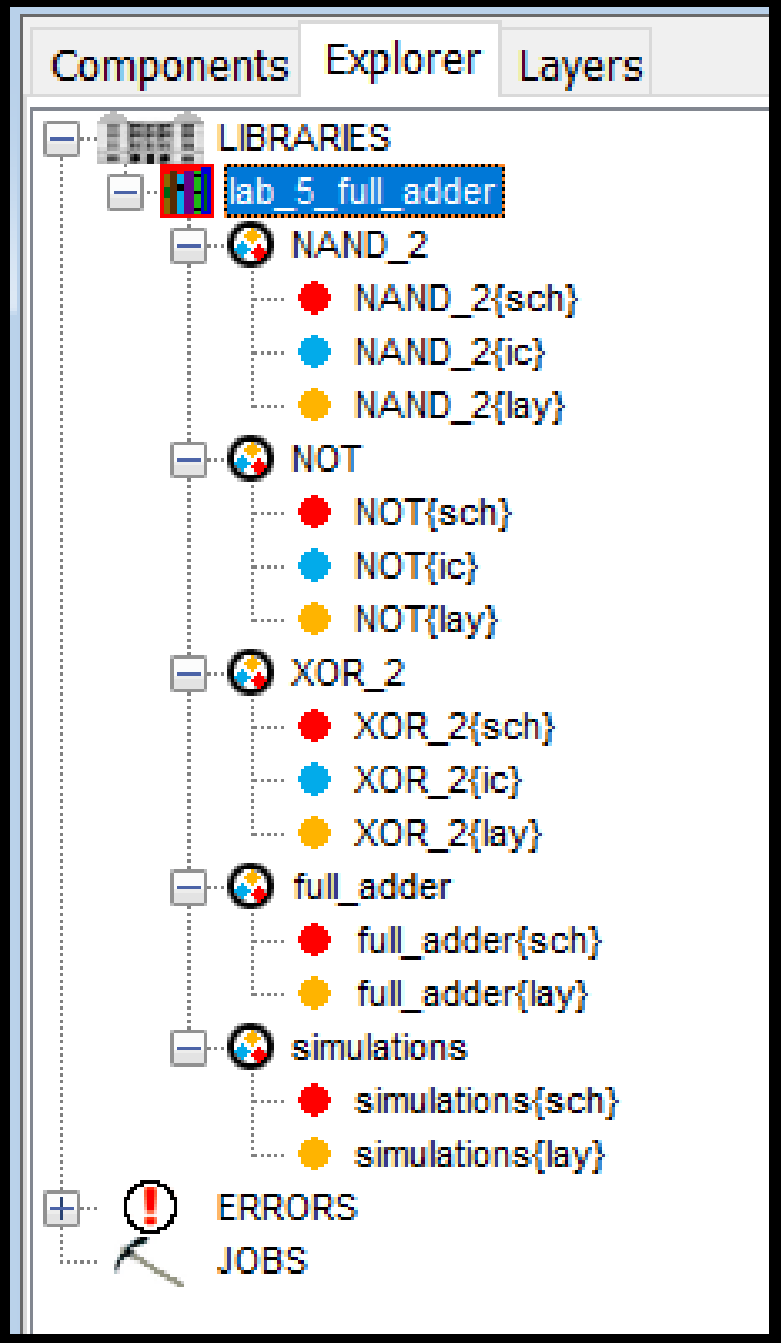
Lab 5

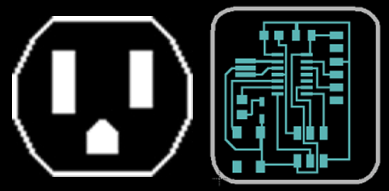
Full Adder

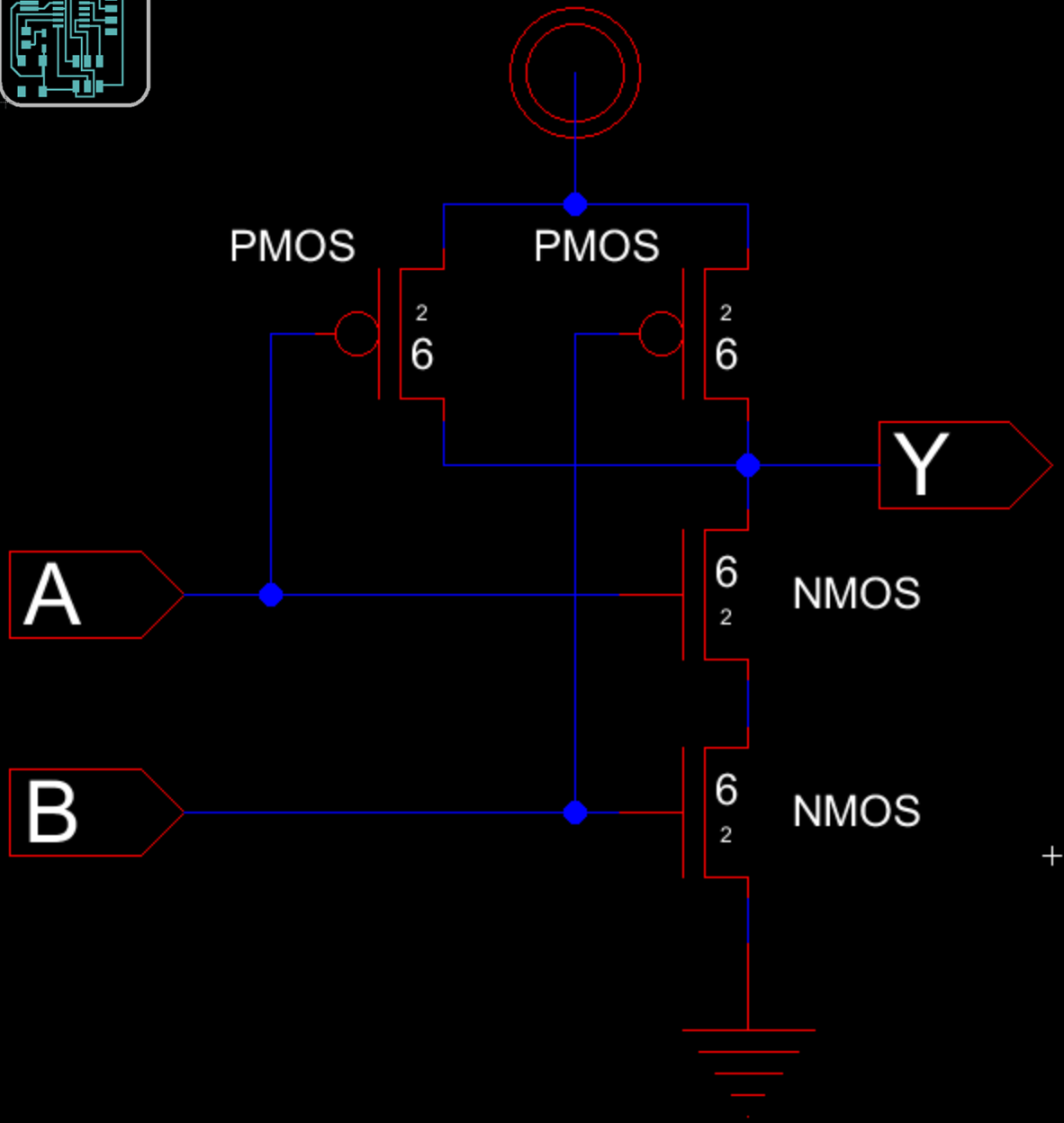
Directions

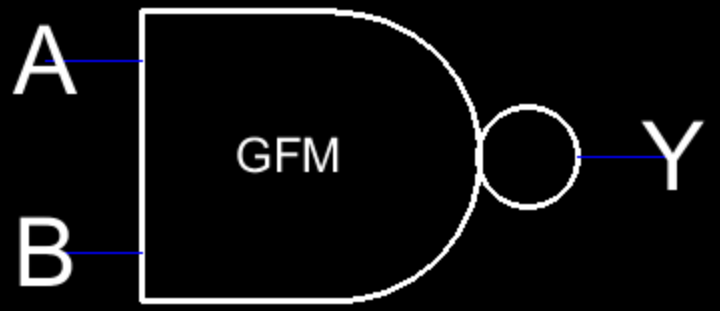
1. Part 1
    - NAND (schematic, layout, simulation)
    - NOT (schematic, layout, simulation)
    - XOR (schematic, layout, simulation)
2. Part 2
    - Full Adder
      (schematic, layout, simulation)

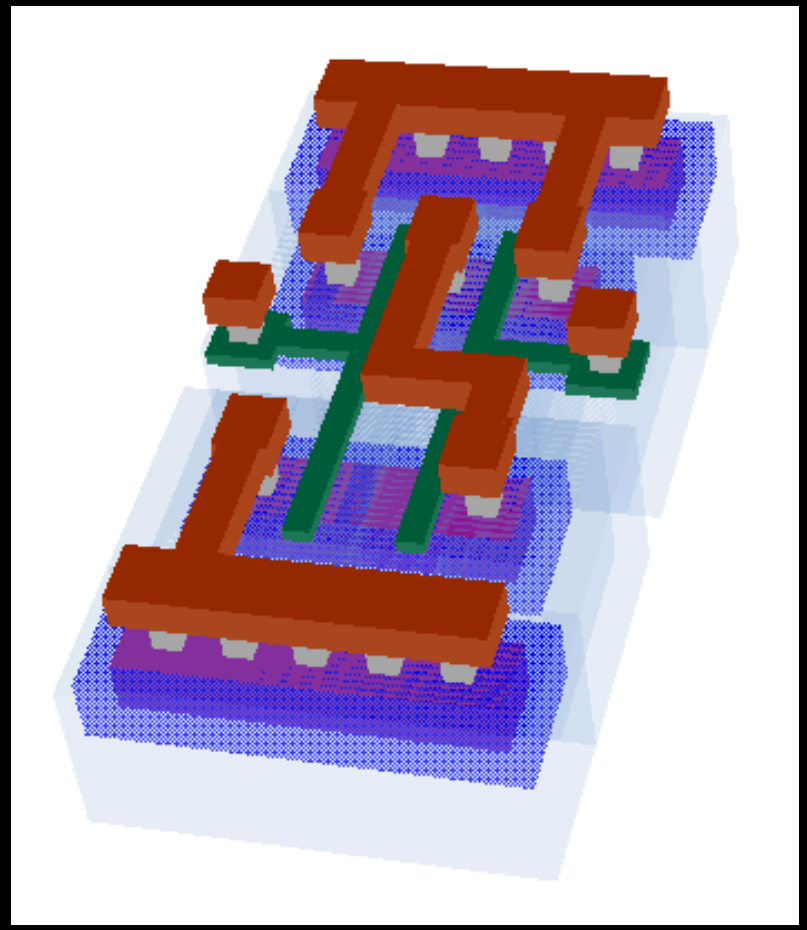

Components
Explorer
Layers
LIBRARIES
lab_5_full_adder
NAND_2
NAND_2{sch}
NAND_2{ic}
NAND_2{lay}
NOT
NOT{sch}
NOT{ic}
NOT{lay}
XOR_2
XOR_2{sch}
XOR_2{ic}
XOR_2{lay}
full_adder
full_adder{sch}
full_adder{lay}
simulations
simulations{sch}
simulations{lay}
ERRORS
JOBS

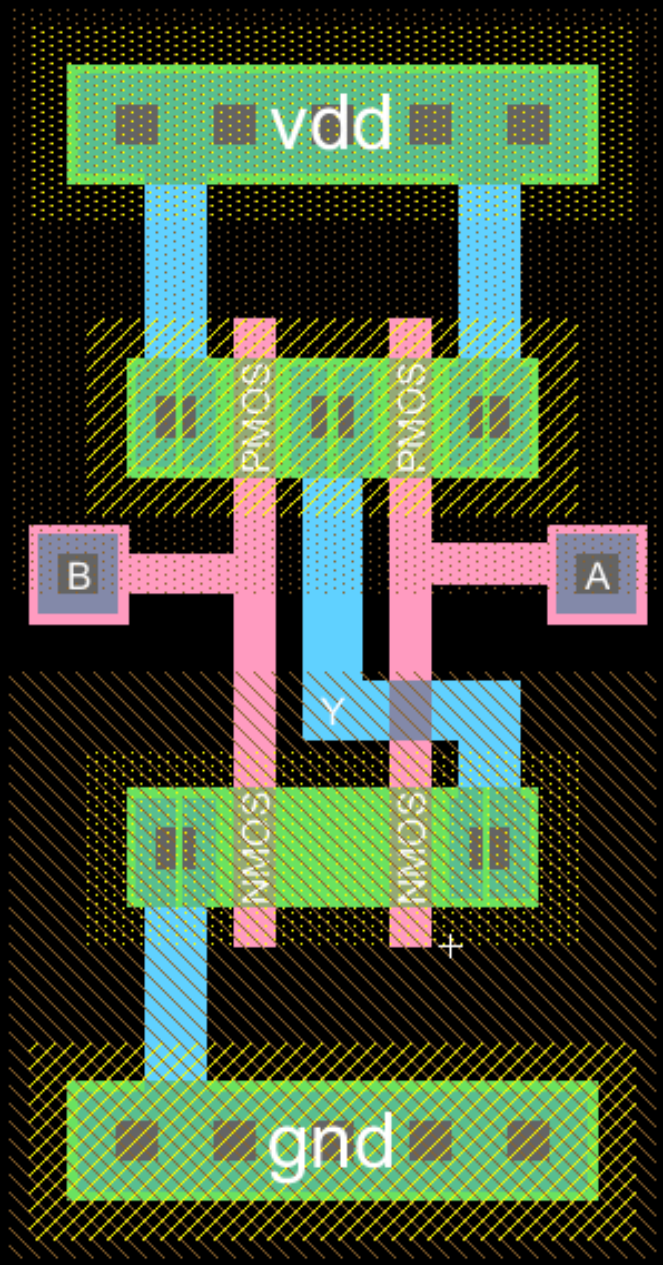
NAND
(schematics, layout, simulations)



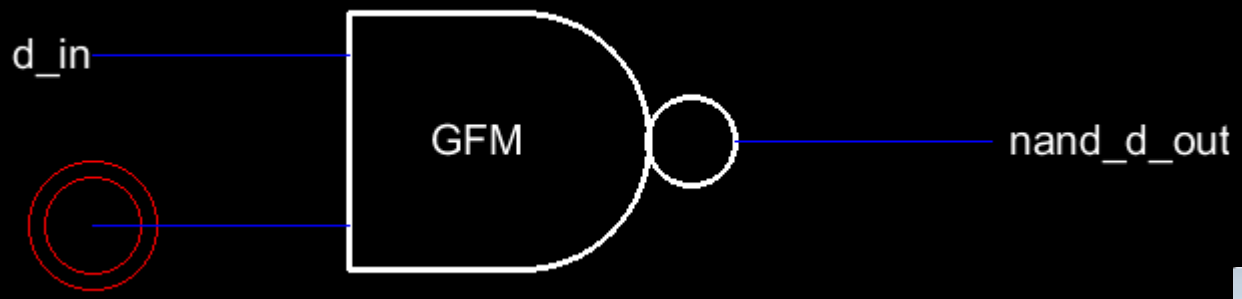

PMOS
PMOS
2
6
2
6
Y
A
6
2
NMOS
B
6
2
NMOS


A
B
GFM
Y



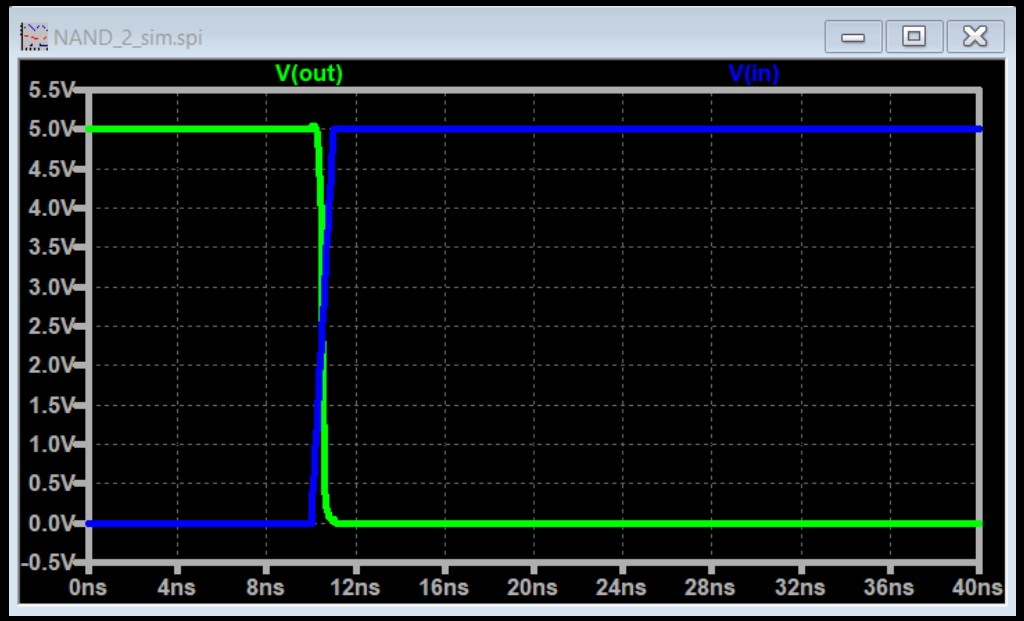

vdd
PMOS
PMOS
B
A
Y
NMOS
NMOS
gnd

Transition Delays




```
vdd vdd 0 dc 5
vin d_in 0 pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n

.include C5_models.txt
```

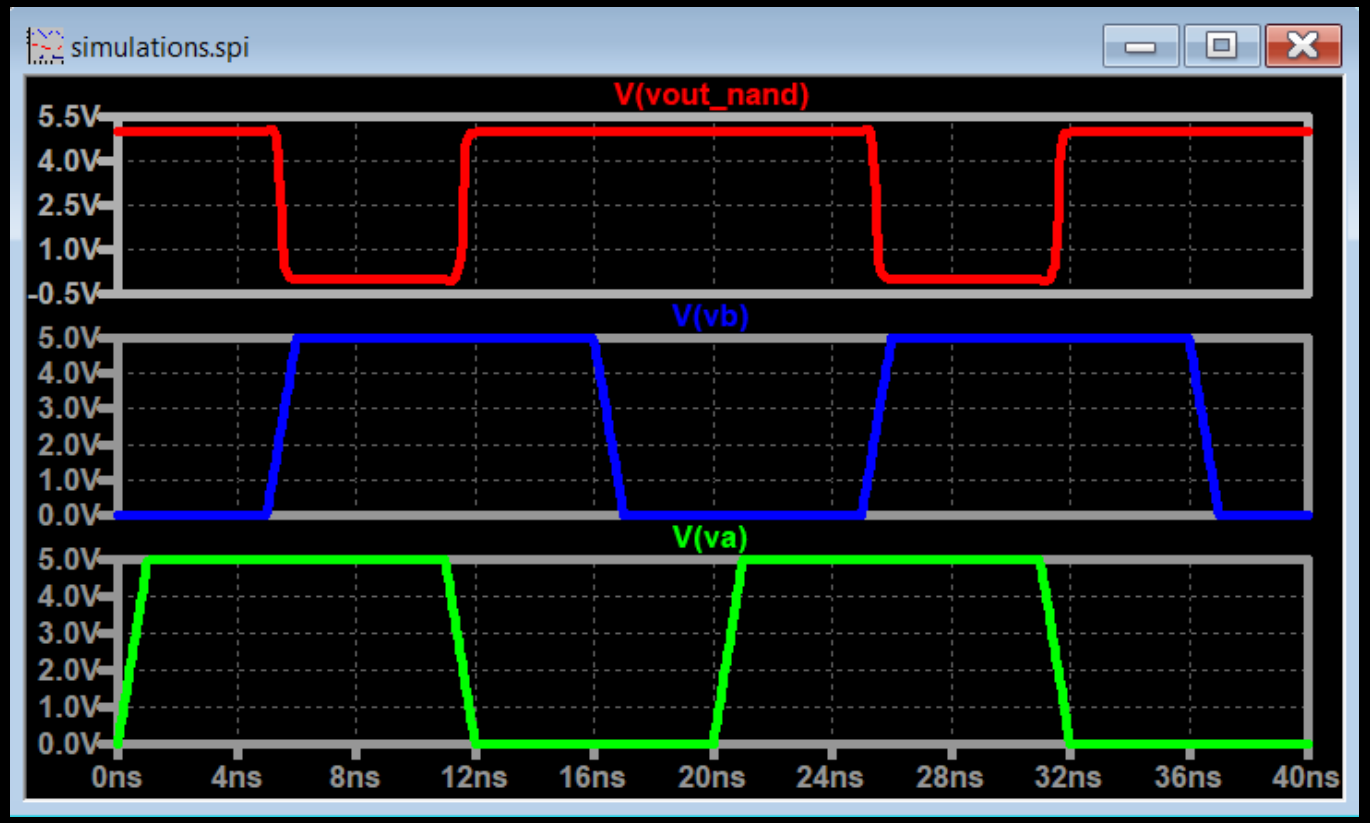


Truth Tables

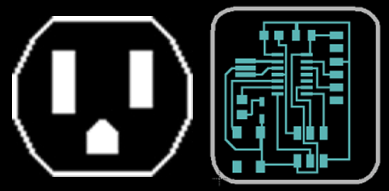


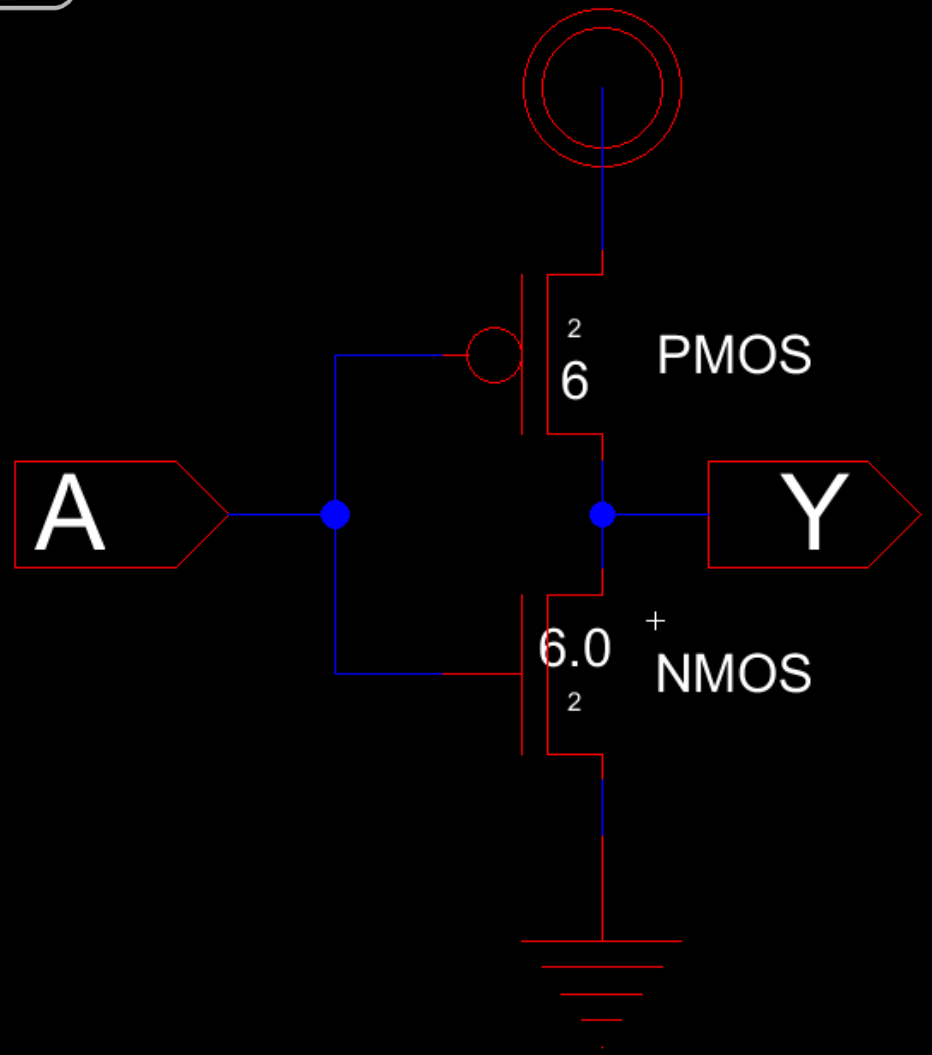

```
vdd vdd 0 dc 5
va va 0 pulse(0v 5v 0n 1n 1n 10n 20n)
vb vb 0 pulse(0v 5v 5n 1n 1n 10n 20n)
.tran 0 40n

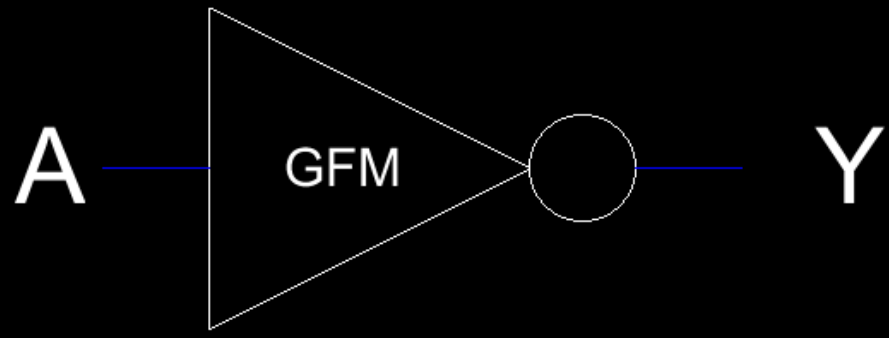
.include C5_models.txt
```

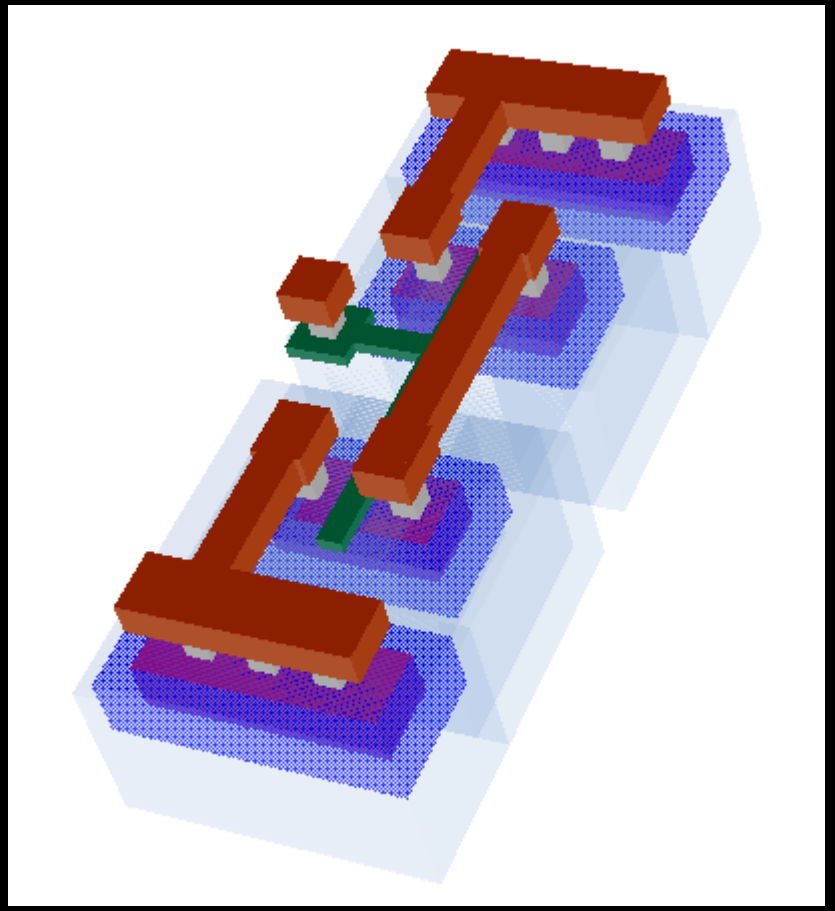


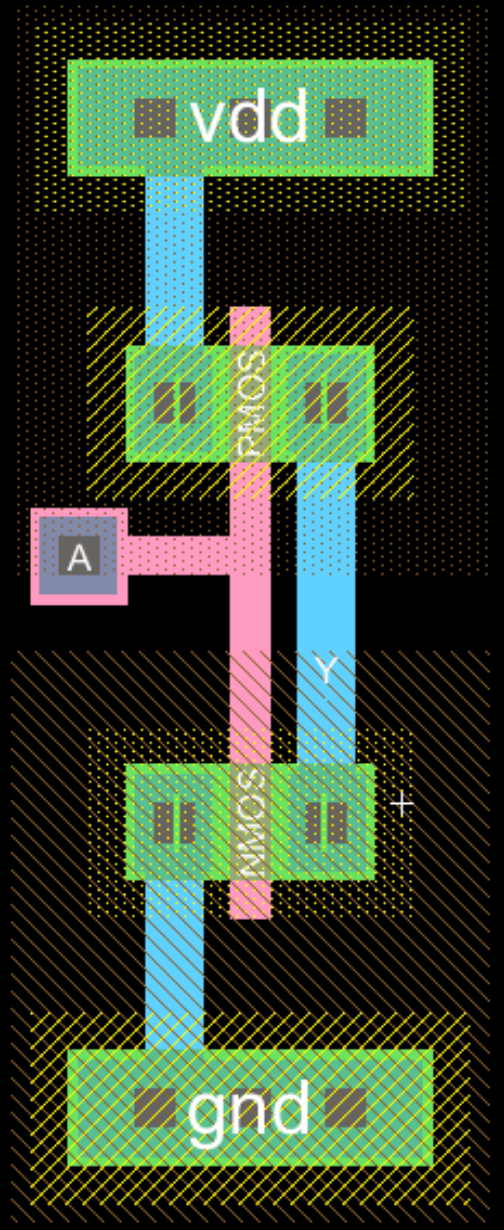
NOT
(schematics, layout, simulations)



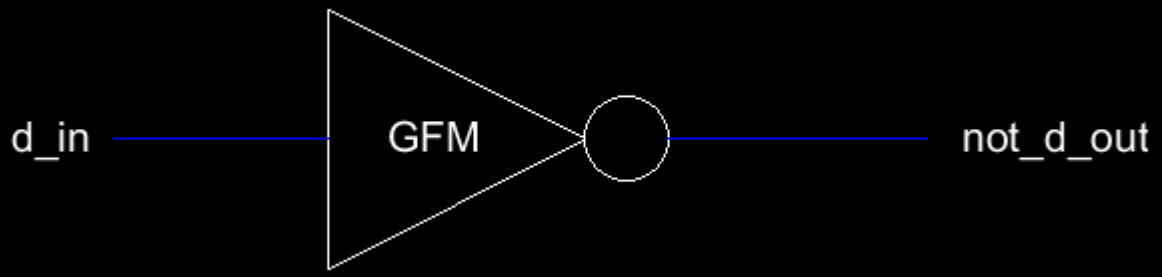

2
6
PMOS
A
Y
6.0
2
NMOS


A
GFM
Y



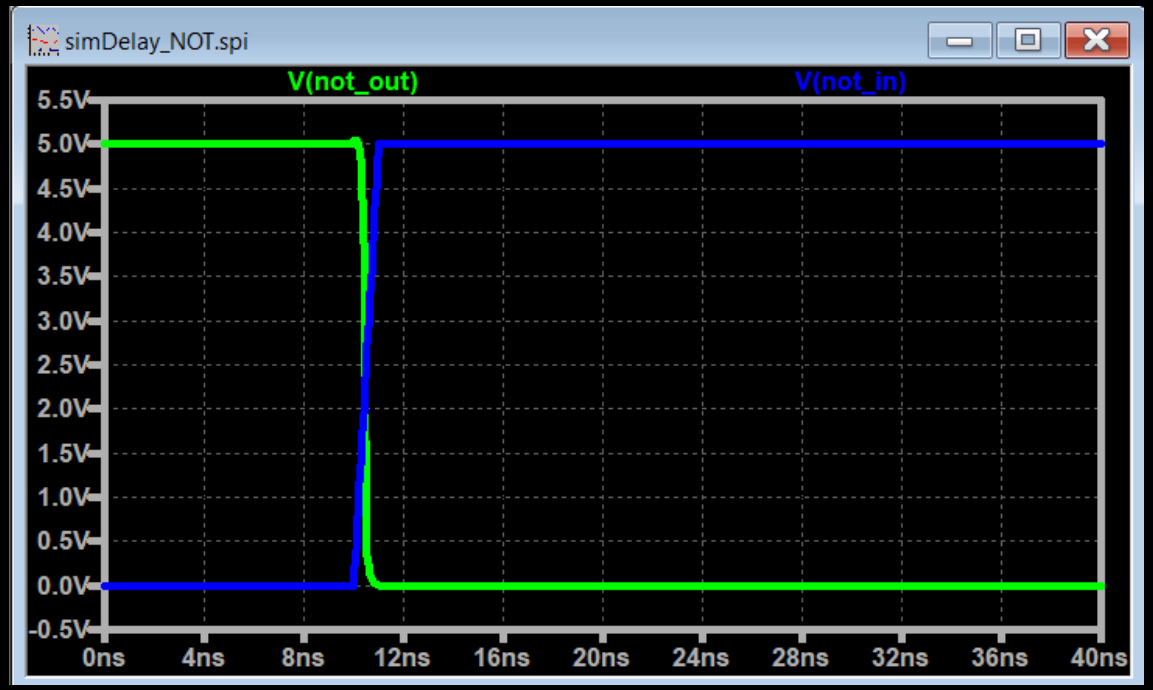

vdd
PMOS
A
Y
NMOS
gnd

Transition Delays

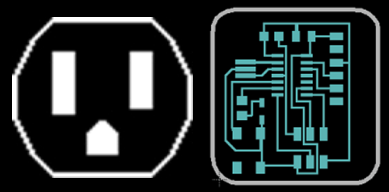



```
vdd vdd 0 dc 5
vin d_in 0 pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n

.include C5_models.txt
```

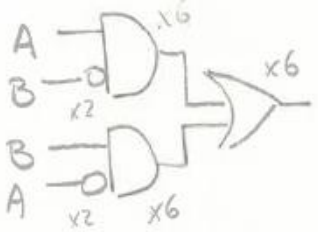


XOR
(schematics, layout, simulations)



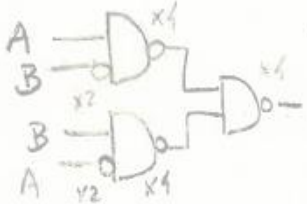
XOR SCHEMATIC

$y = A\bar{B} + B\bar{A}$



22 TRANSISTORS!

MAKE IT ALL NANDS



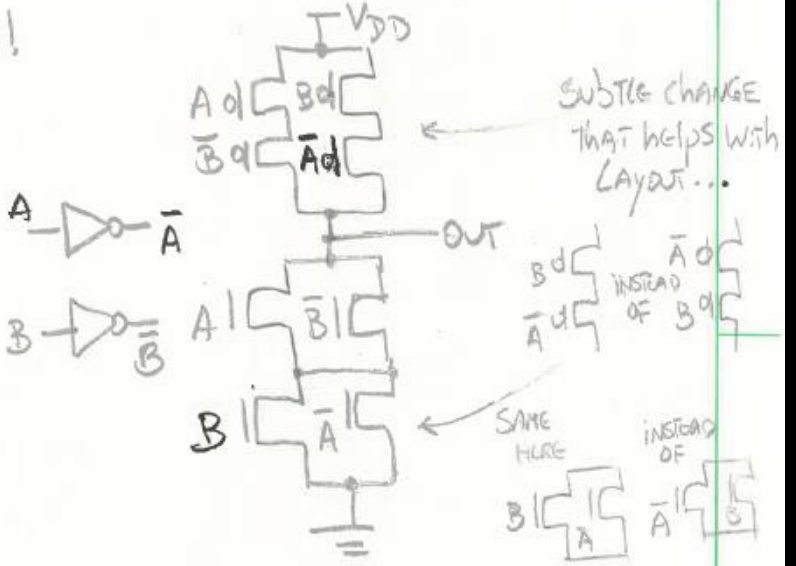
16 TRANSISTORS!

BETTER DESIGN (AOI LOGIC)

$\bar{y} = \overline{A\bar{B} + B\bar{A}}$

$\bar{y} = (\bar{A}+B)(\bar{B}+A)$

$y = \overline{(\bar{A}+B)(\bar{B}+A)}$



12 TRANSISTORS (GOOD COMPROMISE)

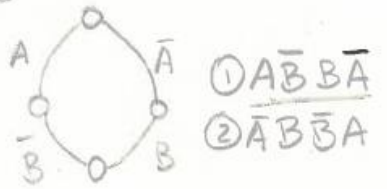
With Less Transistor

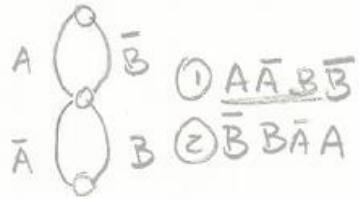
- With pass transistor – 6
- Complete transmission gates – 10 (not pure CMOS implementation)

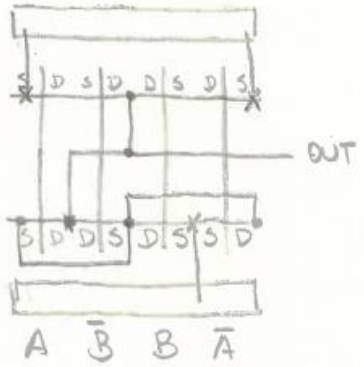
Problem with some voltages

STICK DIAGRAMS

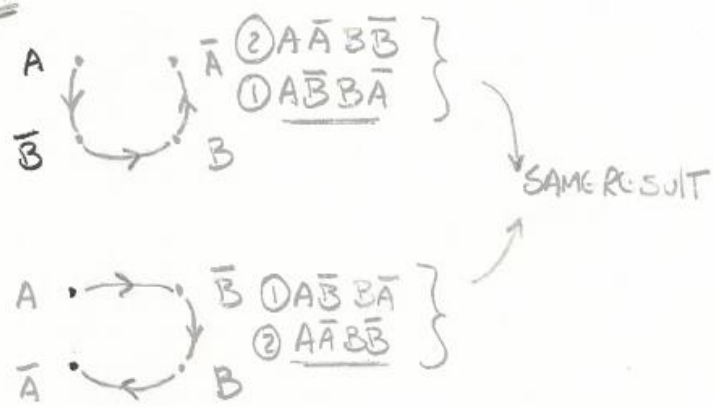
METHOD 1

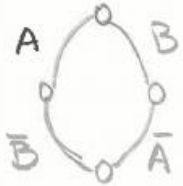




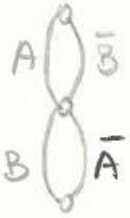


METHOD 2

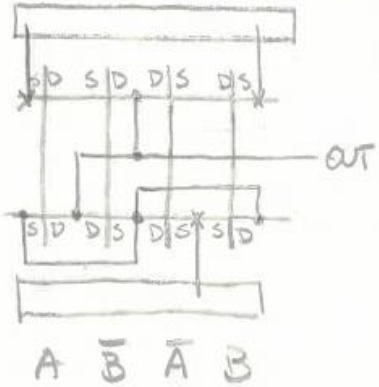


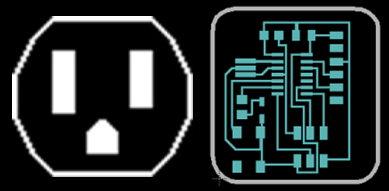


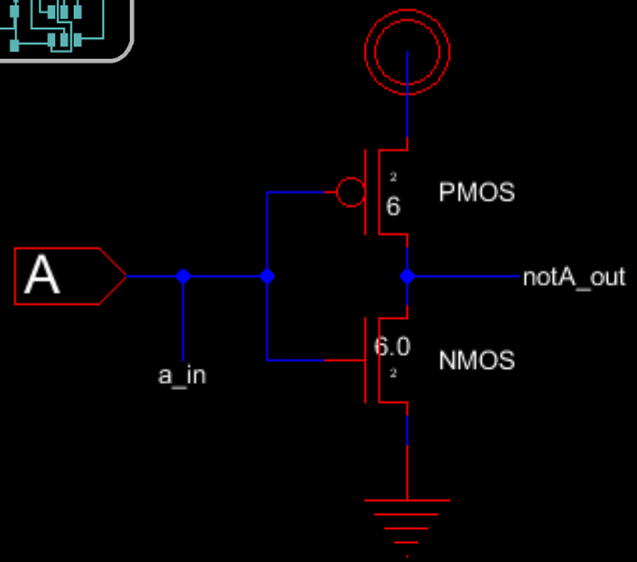
AB̄ĀB

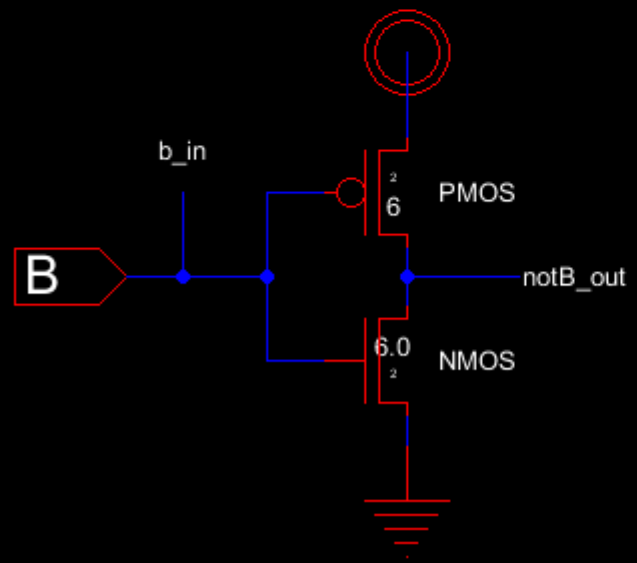


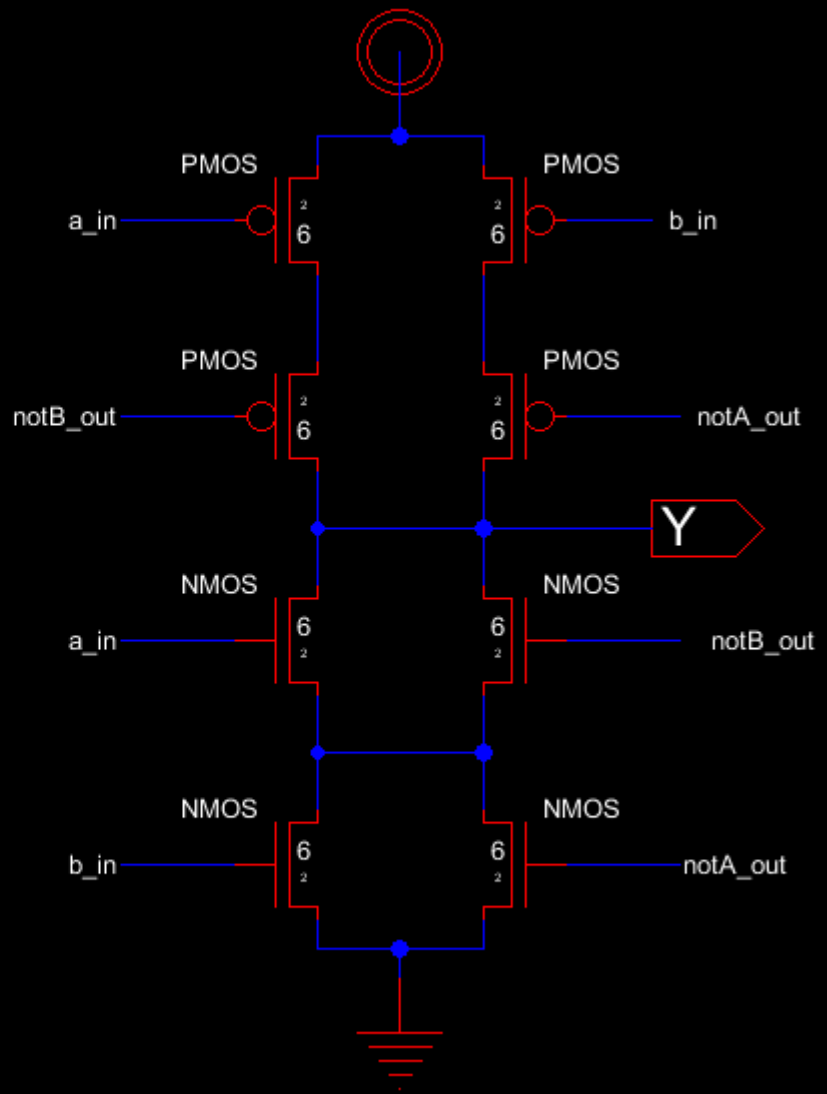
ABĀB̄



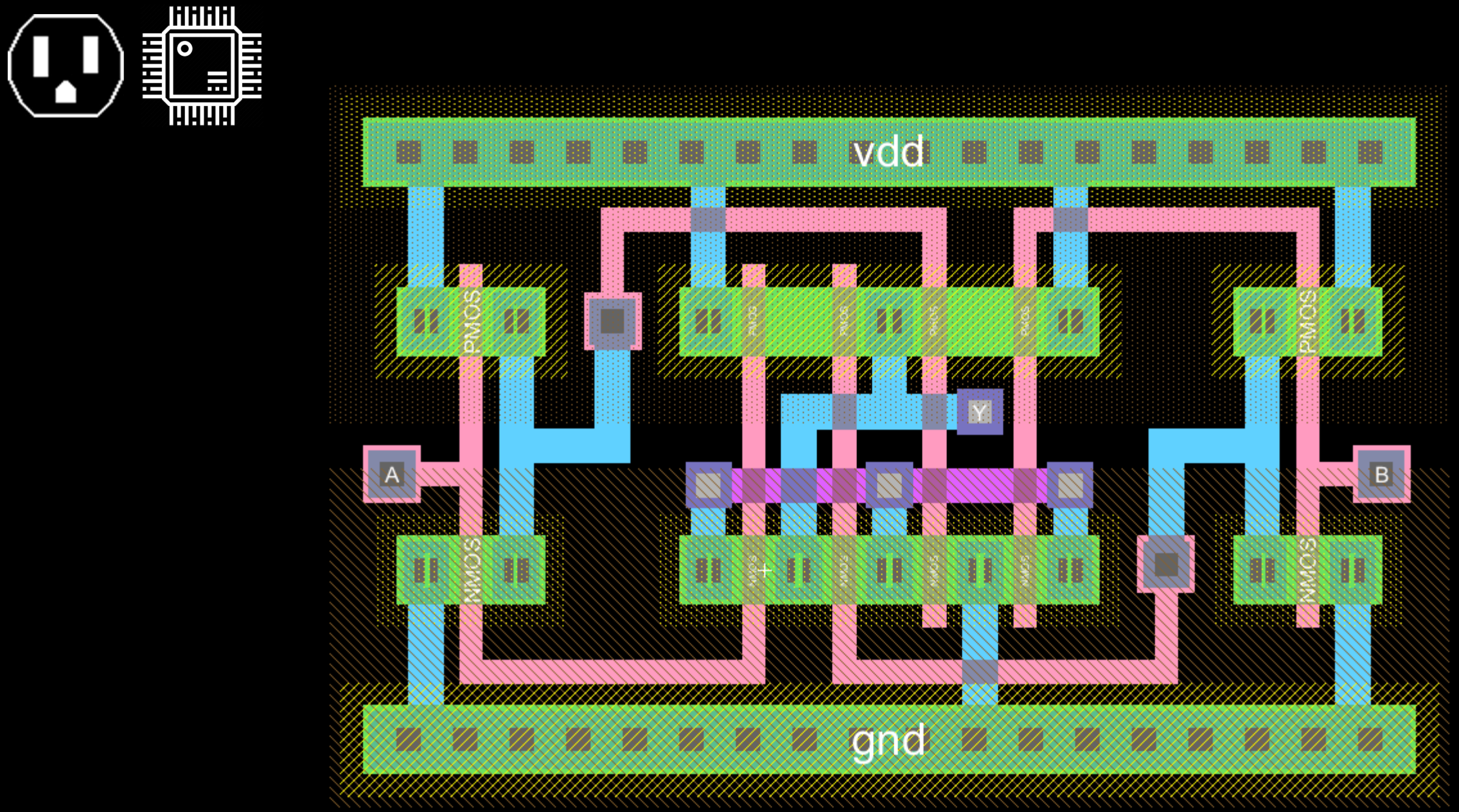


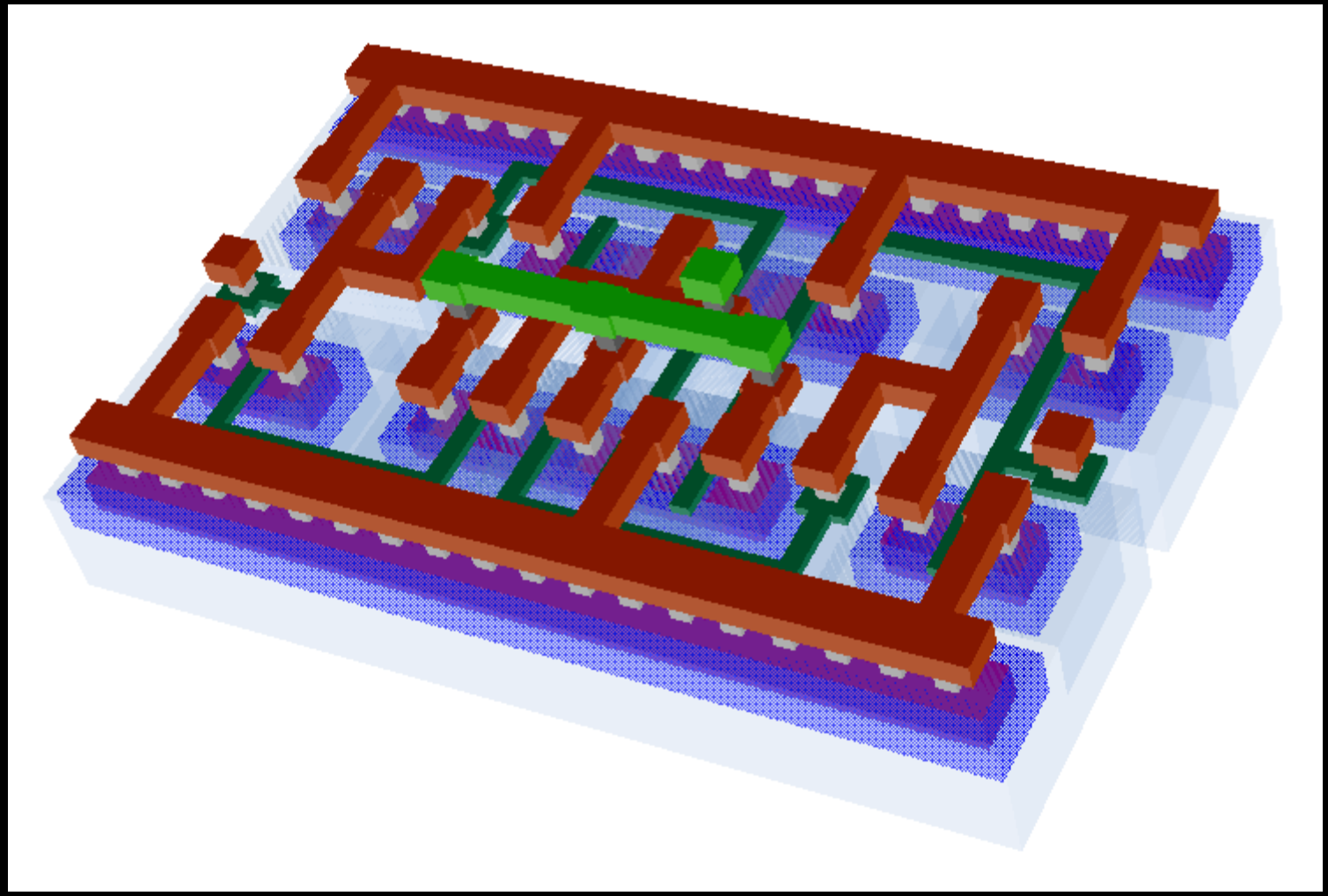

A
a_in
PMOS
notA_out
NMOS
6.0


b_in
B
PMOS
notB_out
NMOS
6.0


PMOS
a_in
PMOS
b_in
PMOS
notB_out
PMOS
notA_out
Y
NMOS
a_in
NMOS
notB_out
NMOS
b_in
NMOS
notA_out


A
B
GFM
Y


vdd
gnd
A
B
Y
PMOS
NMOS



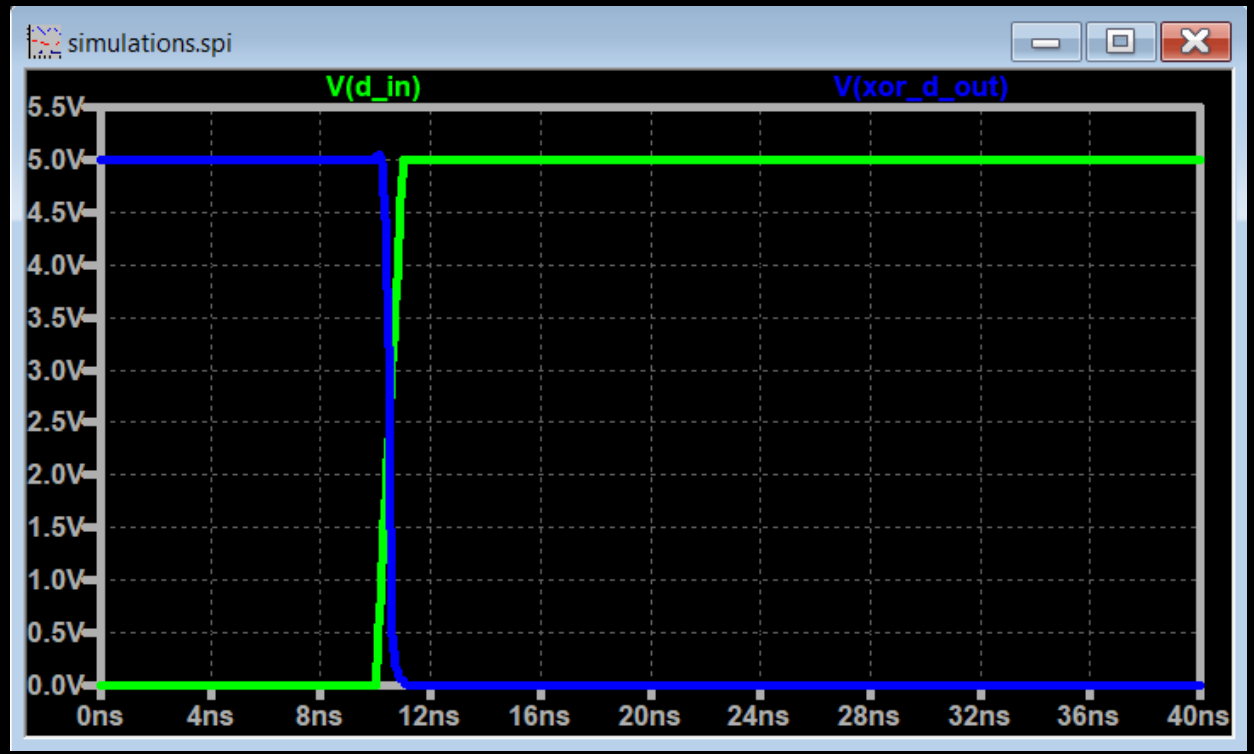
Transition Delays



LTspice®

```
vdd vdd 0 dc 5
vin d_in 0 pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n

.include C5_models.txt
```

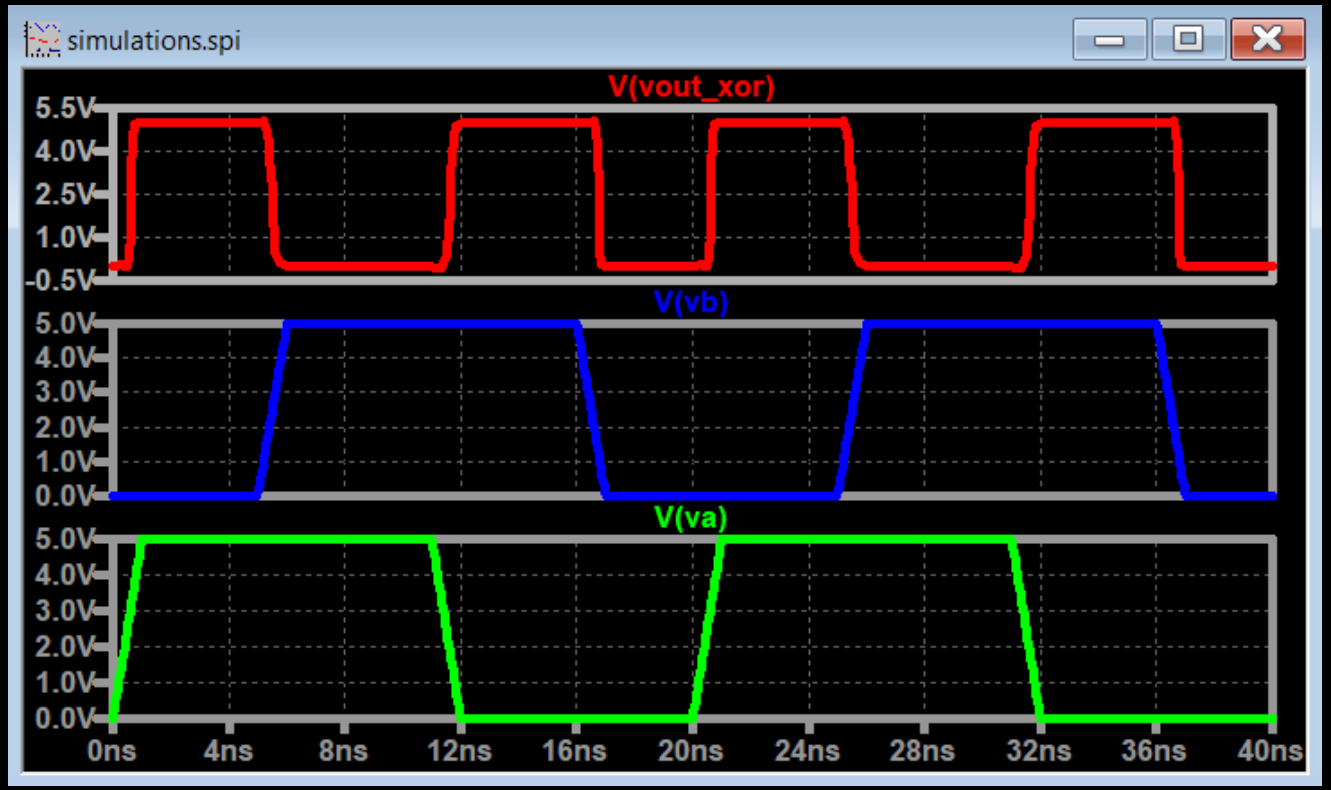


Truth Tables




```
vdd vdd 0 dc 5
va va 0 pulse(0v 5v 0n 1n 1n 10n 20n)
vb vb 0 pulse(0v 5v 5n 1n 1n 10n 20n)
.tran 0 40n
```

.include C5_models.txt



Full Adder

(schematics, layout, simulations)

FULL ADDER

TRUTH TABLE

a	b	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cin \ ab	00	01	11	10
0	0	1	0	1
1	1	0	1	0

Sum(S)

$S = a \oplus b \oplus C_{in}$

Cin \ ab	00	01	11	10
0	0	0	1	0
1	0	1	1	1

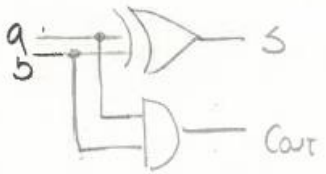
$C_{OUT} = ab + bc_{in} + ac_{in}$

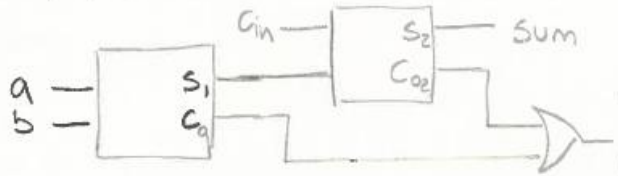
$= ab + c_{in}(a+b)$

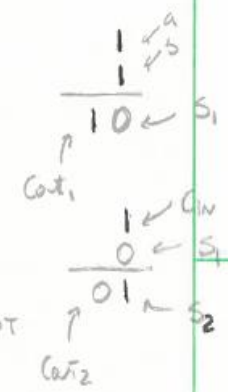


HALF ADDER

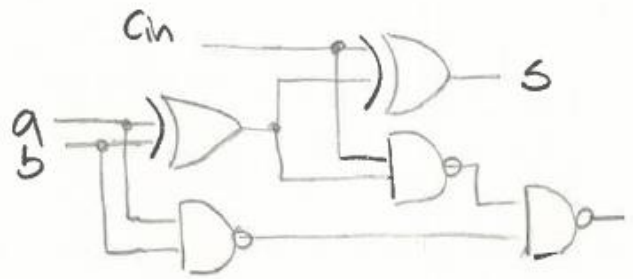
a	b	S	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1







X2 HALF ADDERS USES LESS GATES...



$$\text{SUM} = a \oplus b \oplus C_{in}$$

$$C_{OUT} = ab + C_{in}(a \oplus b)$$

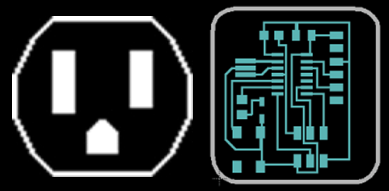
$$= ab + C_{in}(a\bar{b} + \bar{a}b)$$

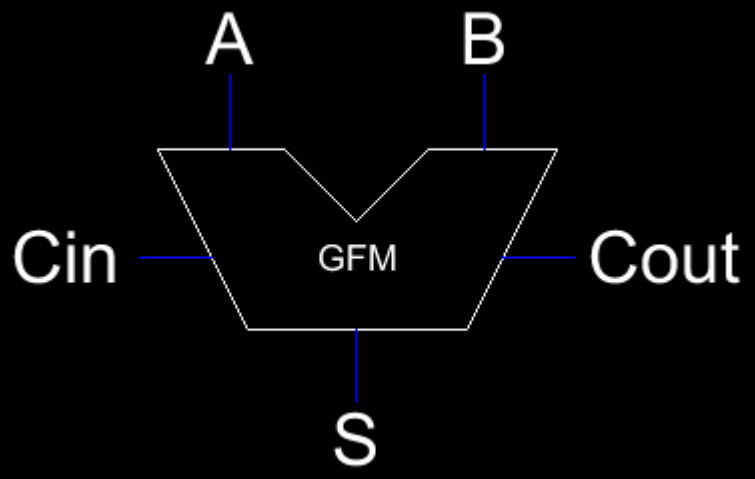
$X + X = X$ $$= ab + C_{in}a\bar{b} + C_{in}\bar{a}b$$

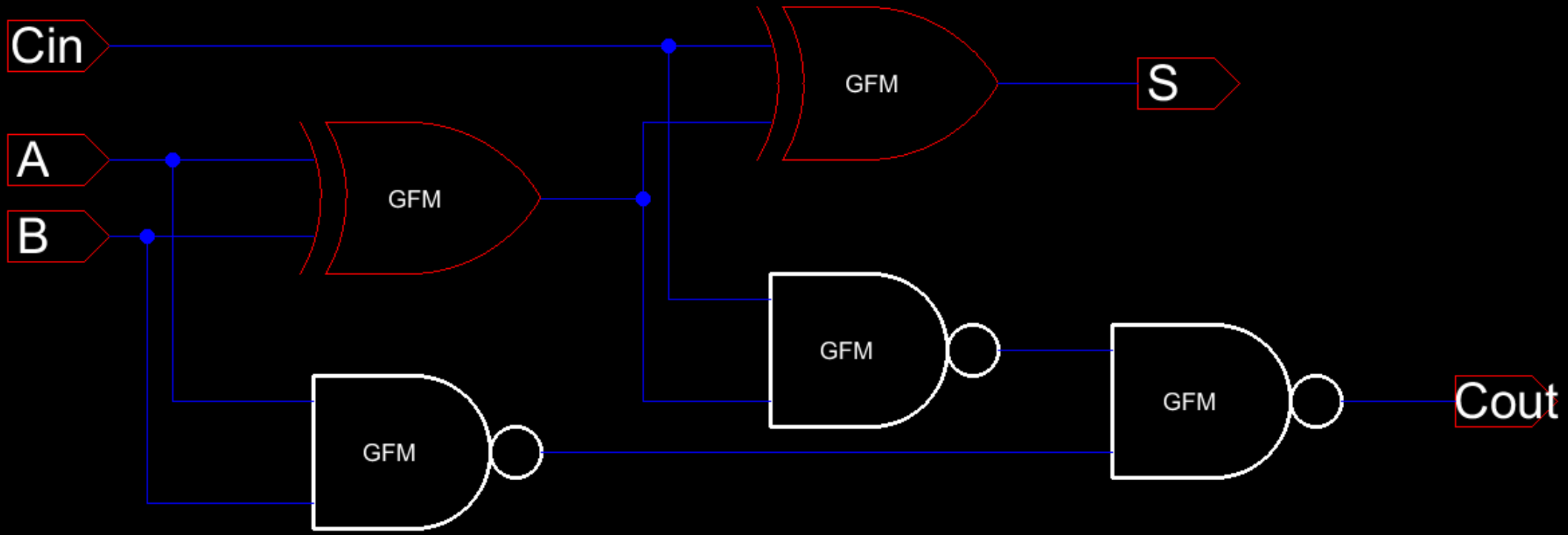
$$= ab + \bar{a}b + C_{in}a\bar{b} + C_{in}\bar{a}b$$

$X + X'y = X + y$ $$= a(b + \bar{b}C_{in}) + b(a + \bar{a}C_{in})$$

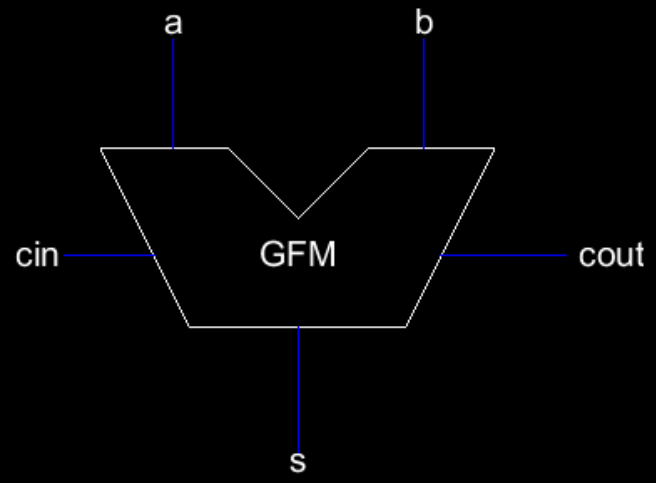
$$= a(b + C_{in}) + b(a + C_{in}) = ab + aC_{in} + ab + bC_{in} = \overline{ab + aC_{in} + bC_{in}}$$




A
B
Cin
GFM
Cout
S

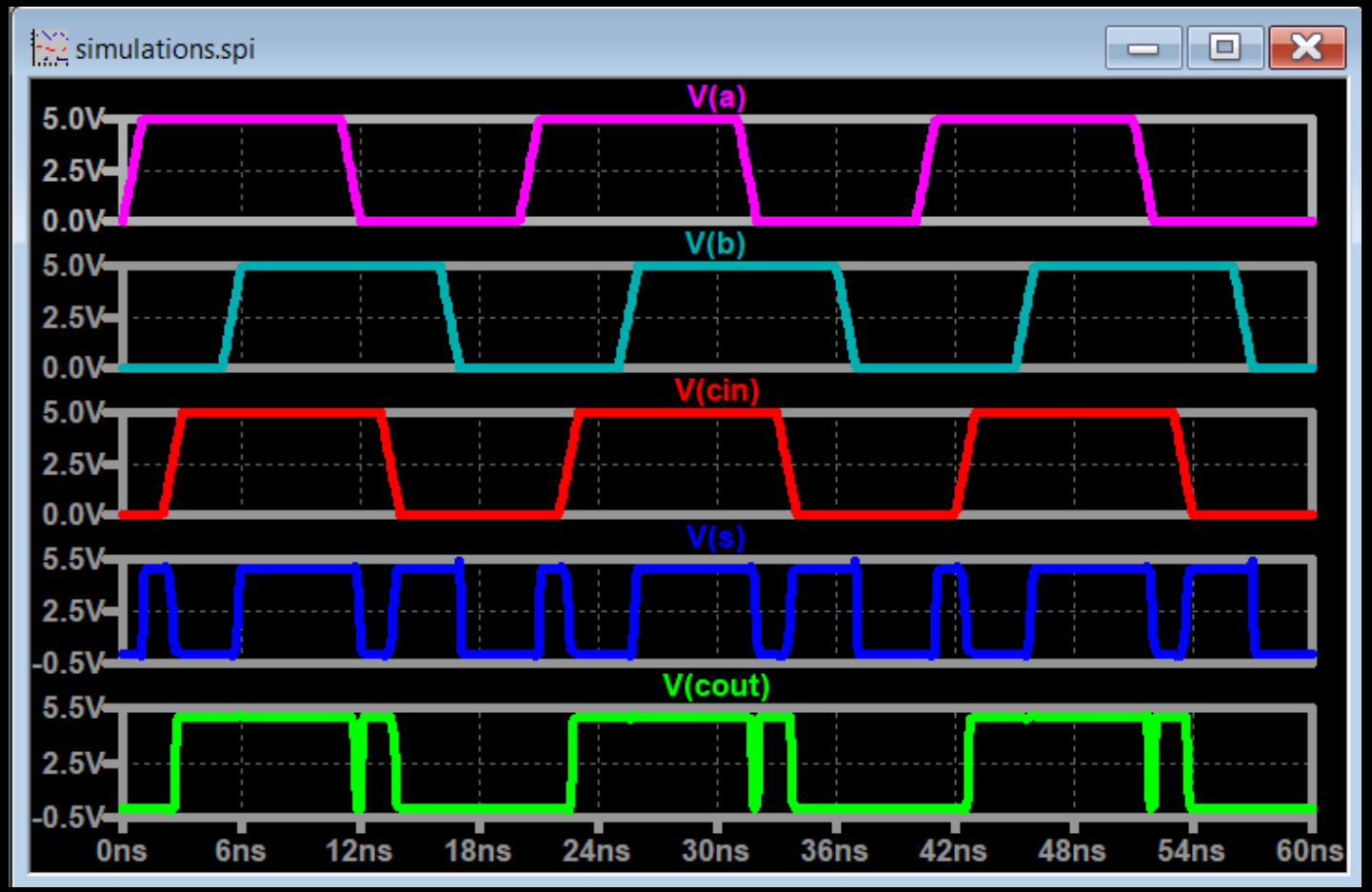

Cin
A
B
GFM
GFM
S
GFM
GFM
GFM
Cout

Truth Tables

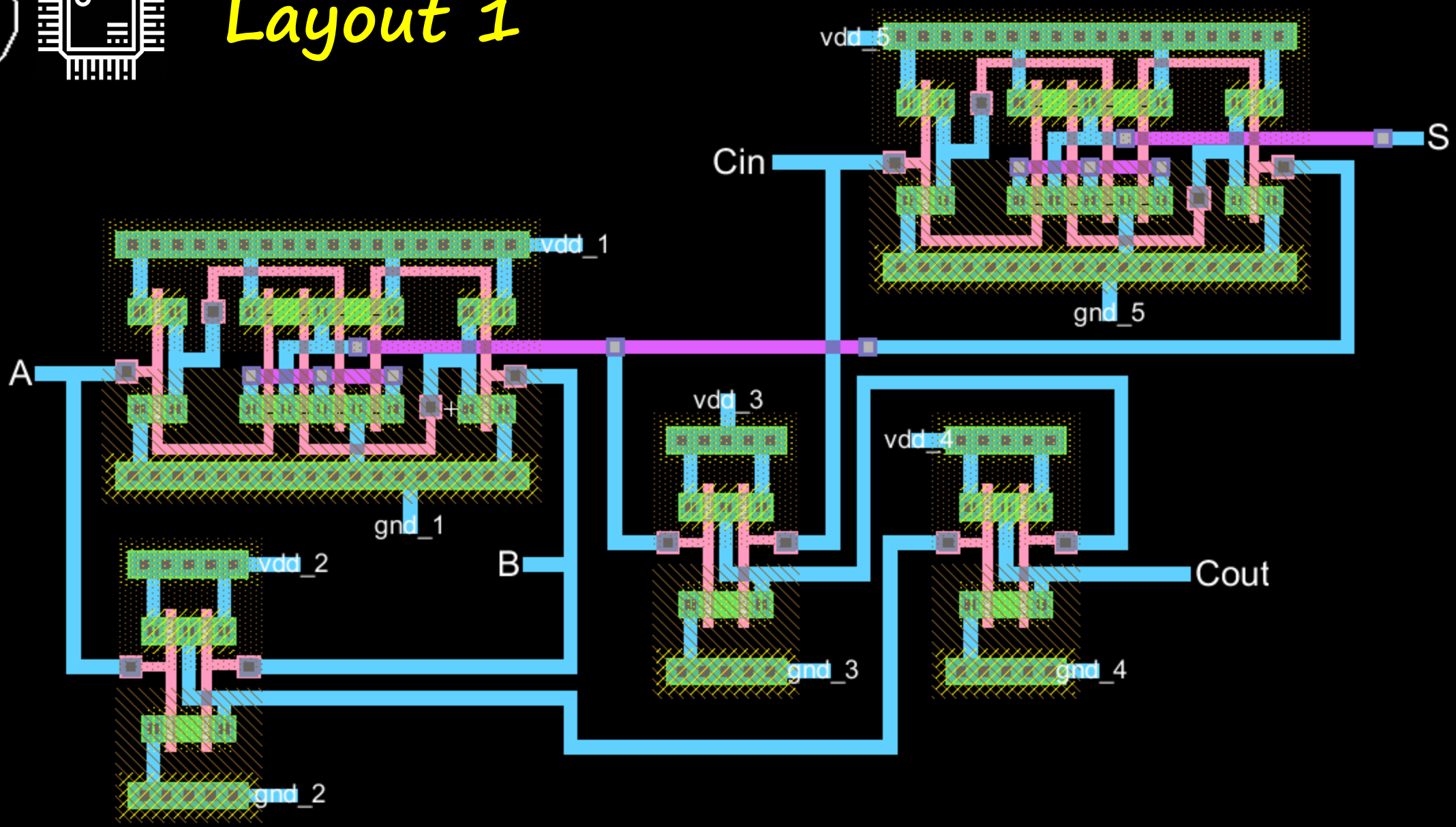


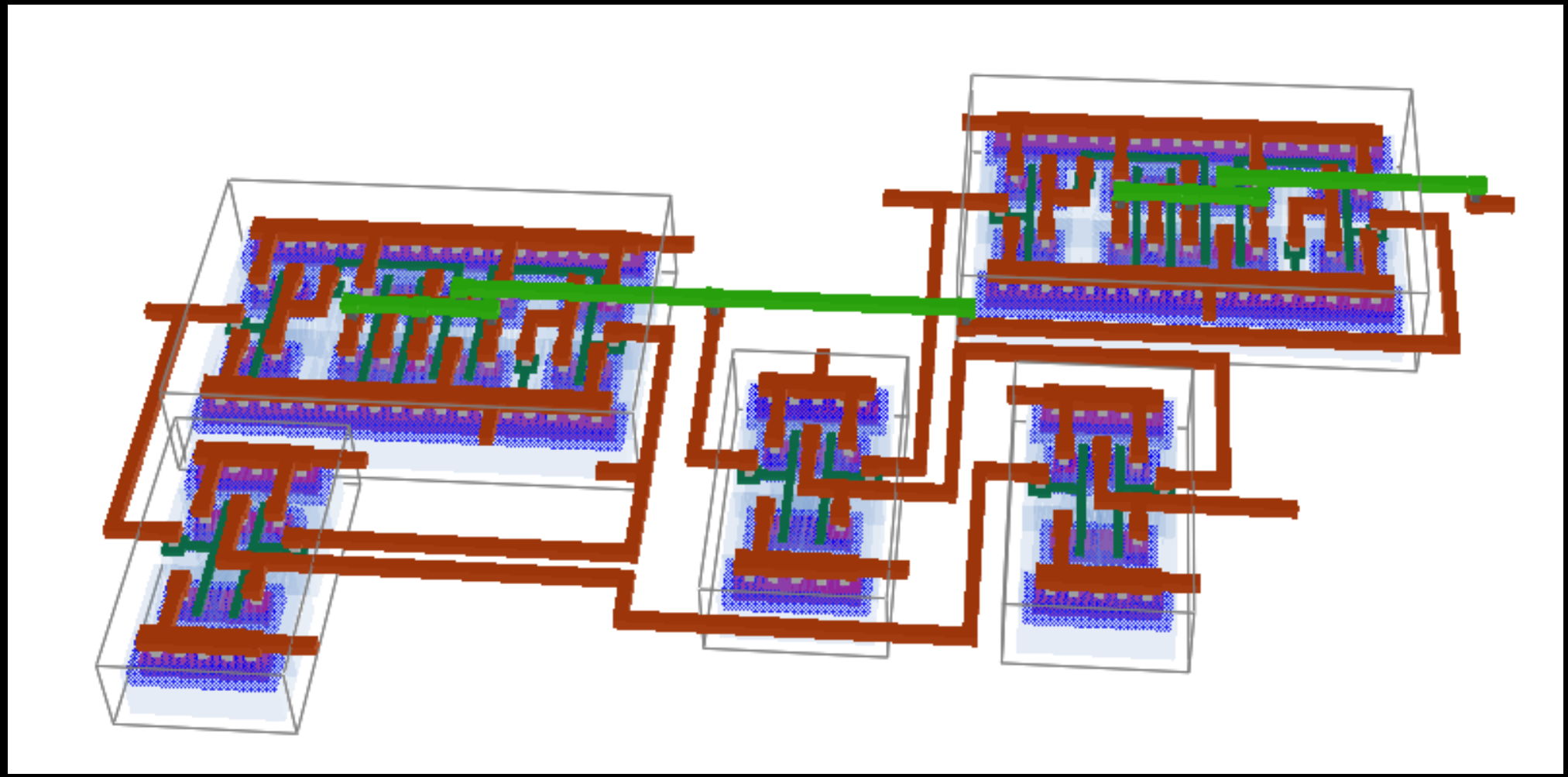

```
vdd vdd 0 dc 5
va a 0 pulse(0v 5v 0n 1n 1n 10n 20n)
vb b 0 pulse(0v 5v 5n 1n 1n 10n 20n)
vcin cin 0 pulse(0v 5v 2n 1n 1n 10n 20n)
.tran 0 60n

.include C5_models.txt
```

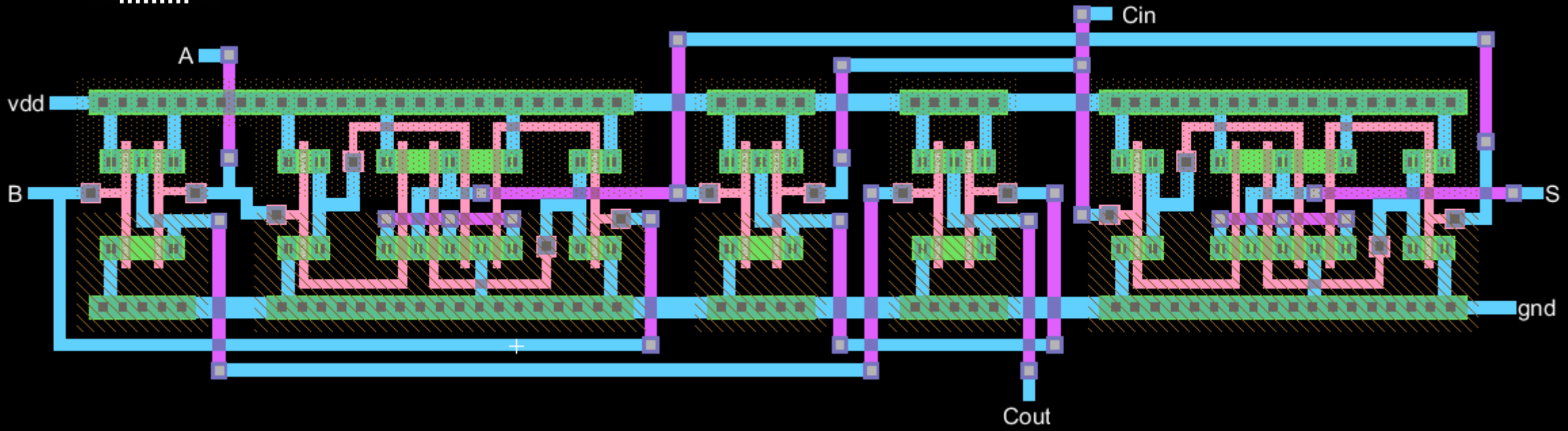


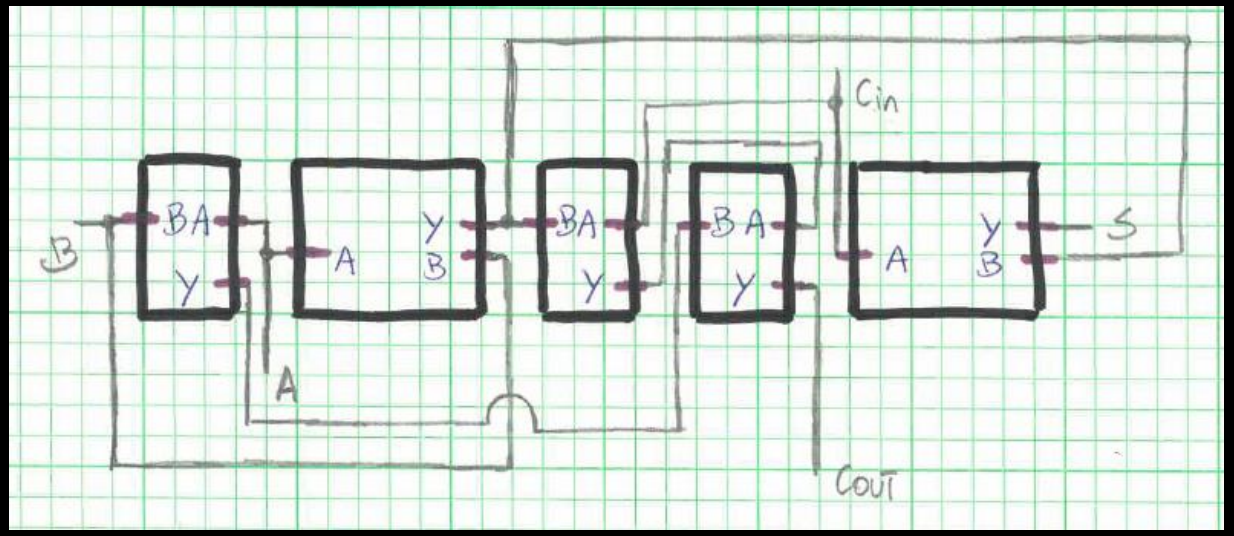
Layout 1

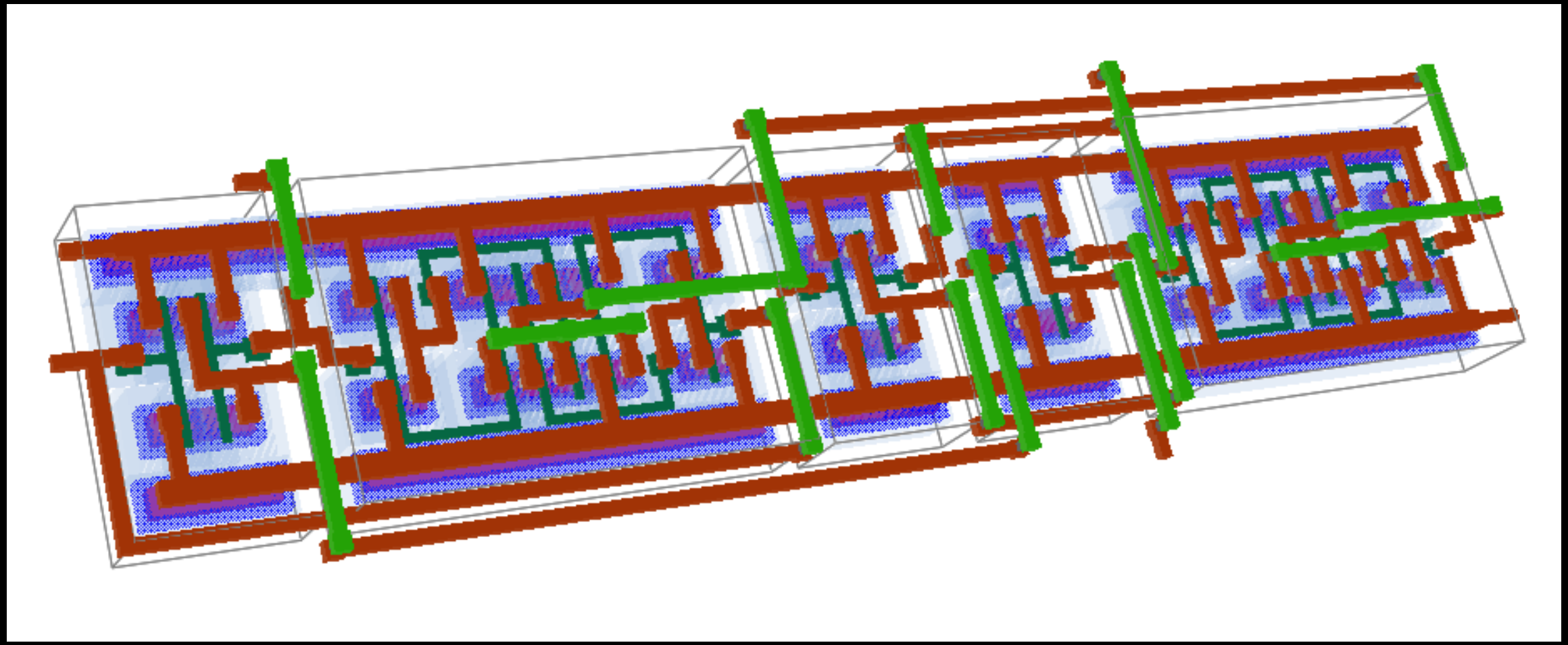

vdd_5
Cin
S
gnd_5
vdd_1
A
gnd_1
vdd_3
vdd_4
vdd_2
B
Cout
gnd_3
gnd_4
gnd_2

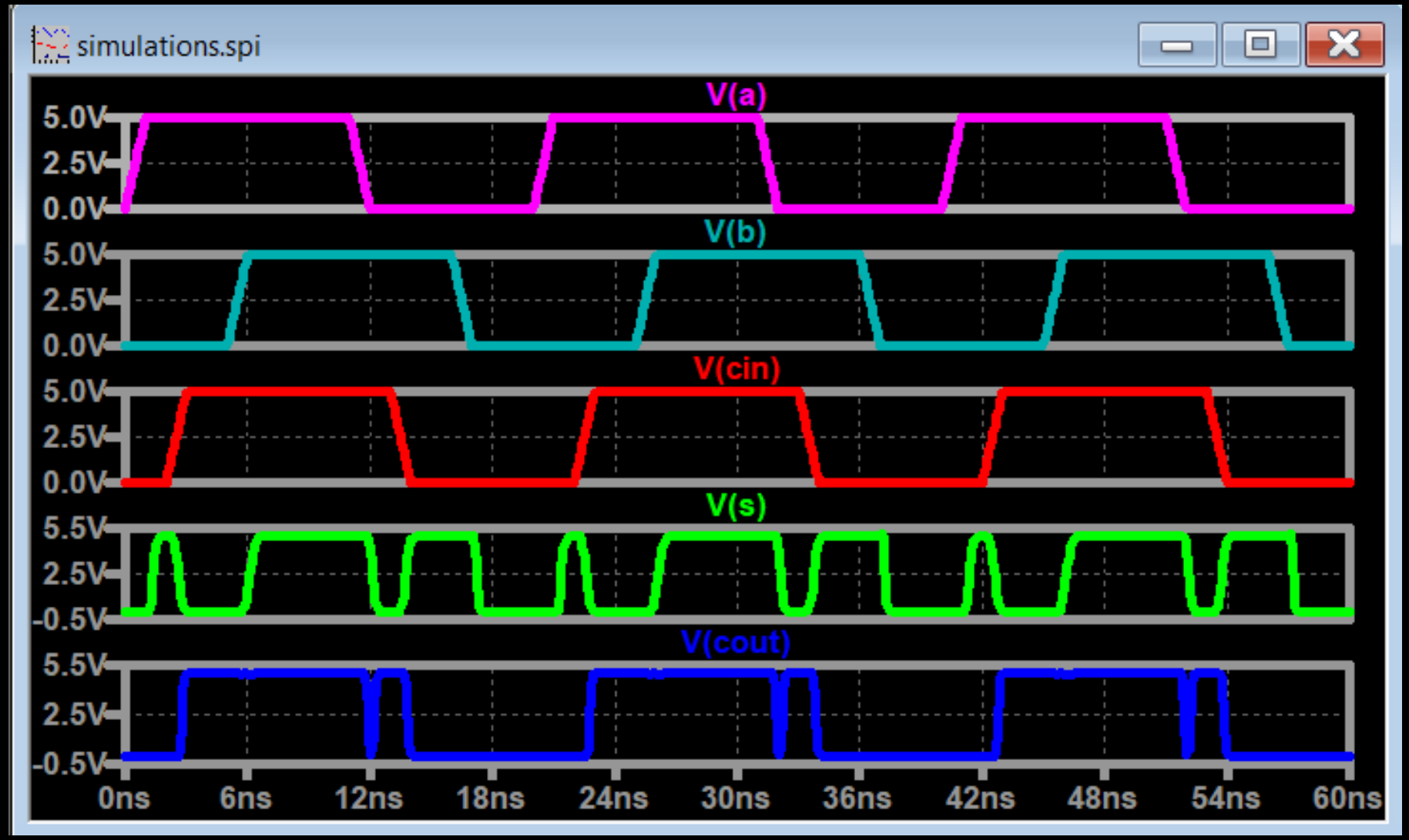


Layout 2








simulations.spi
V(a)
5.0V
2.5V
0.0V
V(b)
5.0V
2.5V
0.0V
V(cin)
5.0V
2.5V
0.0V
V(s)
5.5V
2.5V
-0.5V
V(cout)
5.5V
2.5V
-0.5V
0ns
6ns
12ns
18ns
24ns
30ns
36ns
42ns
48ns
54ns
60ns